CS23532-COMPUTER NETWORKS-LAB MANUAL

Practical -1

AIM: - Study of various Network commands used in Linux and Windows:

Windows commands

C:\Users\Rajesh.B >hostname

Rajesh.B

C:\Users\Rajesh.B >ipconfig /all

Windows IP Configuration

Host Name . . . . . : Rajesh.B

Primary Dns Suffix . . . . . :

Node Type . . . . . : Mixed

IP Routing Enabled. . . . . : No

WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :

Description . . . . . : VirtualBox Host-Only Ethernet Adapter

Physical Address. . . . . : 0A-00-27-00-00-0E

DHCP Enabled. . . . . : No

Autoconfiguration Enabled . . . . : Yes

Link-local IPv6 Address . . . . . : fe80::aa4a:775:1de1:1757%14(Preferred)

IPv4 Address. . . . . : 192.168.56.1(Preferred)

Subnet Mask . . . . . : 255.255.255.0

Default Gateway . . . . . :

DHCPv6 IAID . . . . . : 118095911

DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F

NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected

Connection-specific DNS Suffix . :

Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter

Physical Address. . . . . : F8-FE-5E-3A-70-56

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected

Connection-specific DNS Suffix . :

Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2

Physical Address. . . . . : FA-FE-5E-3A-70-55

DHCP Enabled. . . . . : No

Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :

Description . . . . . : Intel(R) Wi-Fi 6E AX211 160MHz

Physical Address. . . . . : F8-FE-5E-3A-70-55

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

Link-local IPv6 Address . . . . . : fe80::b810:29c9:55e9:3a65%17(Preferred)

IPv4 Address. . . . . : 10.83.96.83(Preferred)

Subnet Mask . . . . . : 255.255.255.0

Lease Obtained. . . . . : 26 August 2026 14:38:14

Lease Expires . . . . . : 26 August 2026 15:38:11

Default Gateway . . . . . : 10.83.96.20

DHCP Server . . . . . : 10.83.96.20

DHCPv6 IAID . . . . . : 167312990

DHCPv6 Client DUID. . . . . : 00-01-00-01-2D-79-FD-43-2C-58-B9-DE-5F-5F

DNS Servers . . . . . : 10.83.96.20

NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected

Connection-specific DNS Suffix . :

Description . . . . . : Bluetooth Device (Personal Area Network)

Physical Address. . . . . : F8-FE-5E-3A-70-59

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected

Connection-specific DNS Suffix . :

Description . . . . . : Realtek Gaming GbE Family Controller

Physical Address. . . . . : 2C-58-B9-DE-5F-5F

DHCP Enabled. . . . . : Yes

Autoconfiguration Enabled . . . . : Yes

C:\Users\Rajesh.B >arp -a

Interface: 192.168.56.1 --- 0xe

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 10.83.96.83 --- 0x11

Internet Address	Physical Address	Type
10.83.96.20	4e-ec-85-ec-ff-15	dynamic
10.83.96.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

C:\Users\Rajesh.B >netstat -r

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Interface List

14...0a 00 27 00 00 0e ......VirtualBox Host-Only Ethernet Adapter

11...f8 fe 5e 3a 70 56 ......Microsoft Wi-Fi Direct Virtual Adapter

4...fa fe 5e 3a 70 55 ......Microsoft Wi-Fi Direct Virtual Adapter #2

17...f8 fe 5e 3a 70 55 ......Intel(R) Wi-Fi 6E AX211 160MHz

19...f8 fe 5e 3a 70 59 ......Bluetooth Device (Personal Area Network)

2...2c 58 b9 de 5f 5f ......Realtek Gaming GbE Family Controller

1...........................Software Loopback Interface 1

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IPv4 Route Table

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Active Routes:

Network Destination	Netmask	Gateway	Interface	Metric
0.0.0.0	0.0.0.0	10.83.96.20	10.83.96.83	35
10.83.96.0	255.255.255.0	On-link	10.83.96.83	291
10.83.96.83	255.255.255.255	On-link	10.83.96.83	291
10.83.96.255	255.255.255.255	On-link	10.83.96.83	291
127.0.0.0	255.0.0.0	On-link	127.0.0.1	331
127.0.0.1	255.255.255.255	On-link	127.0.0.1	331
127.255.255.255	255.255.255.255	On-link	127.0.0.1	331
192.168.56.0	255.255.255.0	On-link	192.168.56.1	281
192.168.56.1	255.255.255.255	On-link	192.168.56.1	281
192.168.56.255	255.255.255.255	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	On-link	127.0.0.1	331
224.0.0.0	240.0.0.0	On-link	192.168.56.1	281
224.0.0.0	240.0.0.0	On-link	10.83.96.83	291
255.255.255.255	255.255.255.255	On-link	127.0.0.1	331
255.255.255.255	255.255.255.255	On-link	192.168.56.1	281

255.255.255.255 255.255.255.255   On-link   10.83.96.83   291

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Persistent Routes:

None

IPv6 Route Table

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Active Routes:

If Metric Network Destination   Gateway

1   331 ::1/128   On-link

14   281 fe80::/64   On-link

17   291 fe80::/64   On-link

14   281 fe80::aa4a:775:1de1:1757/128

On-link

17   291 fe80::b810:29c9:55e9:3a65/128

On-link

1   331 ff00::/8   On-link

14   281 ff00::/8   On-link

17   291 ff00::/8   On-link

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Persistent Routes:

None

C:\Users\Rajesh.B   >nslookup www.google.com

Server: UnKnown

Address: 10.83.96.20

Non-authoritative answer:

Name: www.google.com

Addresses: 2001:4860:482d:7700::

2001:4860:4826:7700::

2001:4860:4828:7700::

2001:4860:482b:7700::

2001:4860:4827:7700::

2001:4860:4829:7700::

2001:4860:482a:7700::

2001:4860:482c:7700::

142.251.152.119

142.251.156.119

142.251.154.119

142.251.151.119

142.251.153.119

142.251.157.119

142.251.150.119

142.251.155.119

C:\Users\Rajesh.B >ping google.com

Pinging google.com [142.251.43.46] with 32 bytes of data: Reply

from 142.251.43.46: bytes=32 time=14ms TTL=114

Reply from 142.251.43.46: bytes=32 time=233ms TTL=114

Reply from 142.251.43.46: bytes=32 time=59ms TTL=114

Reply from 142.251.43.46: bytes=32 time=328ms TTL=114

Ping statistics for 142.251.43.46:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 14ms, Maximum = 328ms, Average = 158ms

C:\Users\Rajesh.B >route print

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Interface List

14...0a 00 27 00 00 0e ......VirtualBox Host-Only Ethernet Adapter

11...f8 fe 5e 3a 70 56 ......Microsoft Wi-Fi Direct Virtual Adapter

4...fa fe 5e 3a 70 55 ......Microsoft Wi-Fi Direct Virtual Adapter #2

17...f8 fe 5e 3a 70 55 ......Intel(R) Wi-Fi 6E AX211 160MHz

19...f8 fe 5e 3a 70 59 ......Bluetooth Device (Personal Area Network)

2...2c 58 b9 de 5f 5f ......Realtek Gaming GbE Family Controller

1...........................Software Loopback Interface 1

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IPv4 Route Table

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Active Routes:

Network Destination Netmask Gateway Interface Metric

0.0.0.0 0.0.0.0 10.83.96.20 10.83.96.83 35

10.83.96.0 255.255.255.0 On-link 10.83.96.83 291

10.83.96.83 255.255.255.255 On-link 10.83.96.83 291

10.83.96.255 255.255.255.255 On-link 10.83.96.83 291

127.0.0.0 255.0.0.0 On-link 127.0.0.1 331

127.0.0.1 255.255.255.255 On-link 127.0.0.1 331

127.255.255.255 255.255.255.255 On-link 127.0.0.1 331

192.168.56.0 255.255.255.0 On-link 192.168.56.1 281

```
    192.168.56.1  255.255.255.255        On-link     192.168.56.1    281
  192.168.56.255  255.255.255.255        On-link     192.168.56.1    281
      224.0.0.0       240.0.0.0        On-link        127.0.0.1    331
      224.0.0.0       240.0.0.0        On-link     192.168.56.1    281
      224.0.0.0       240.0.0.0        On-link      10.83.96.83    291
  255.255.255.255  255.255.255.255        On-link        127.0.0.1    331
  255.255.255.255  255.255.255.255        On-link     192.168.56.1    281
  255.255.255.255  255.255.255.255        On-link      10.83.96.83    291
===========================================================================

Persistent Routes:
  None

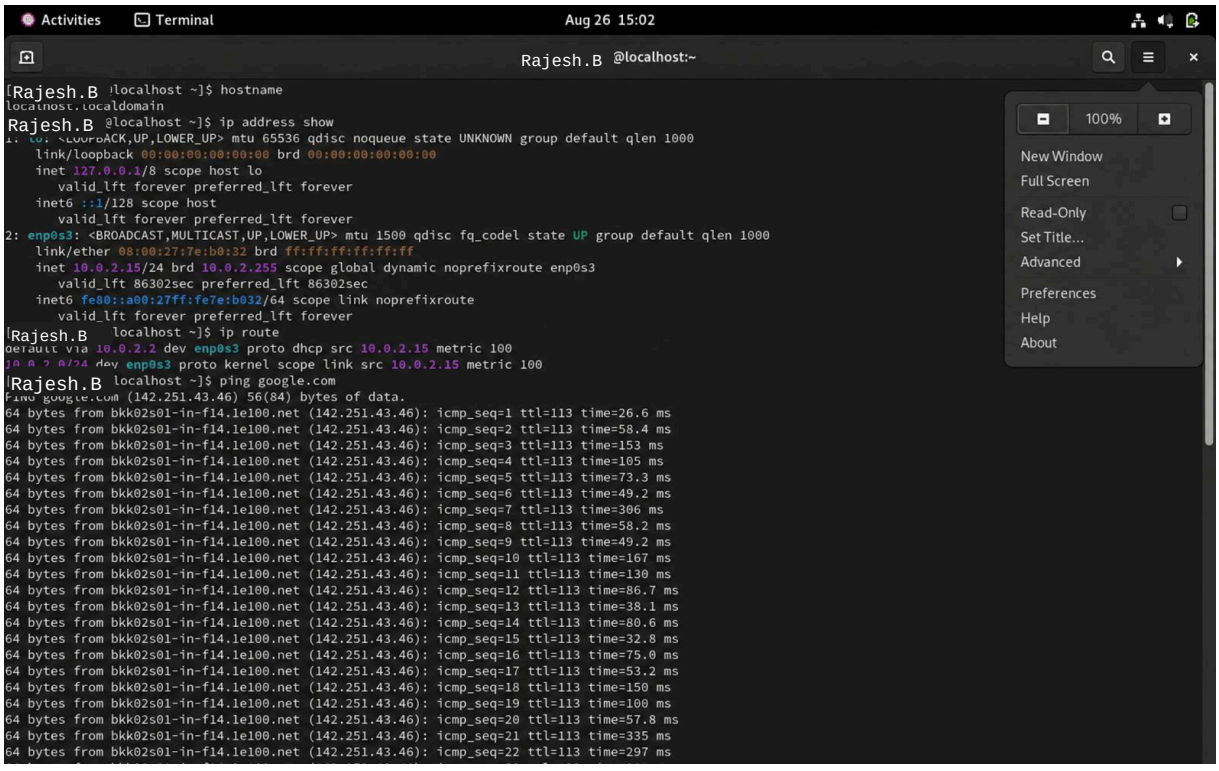

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
 14    281 fe80::/64                On-link
 17    291 fe80::/64                On-link
 14    281 fe80::aa4a:775:1de1:1757/128
                                    On-link
 17    291 fe80::b810:29c9:55e9:3a65/128
                                    On-link
  1    331 ff00::/8                 On-link
 14    281 ff00::/8                 On-link
 17    291 ff00::/8                 On-link
===========================================================================

Persistent Routes:
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None

C:\Users\Rajesh.B >tracert google.com

Linux commands



Student observation answers

1. **Command used to check reachability:** ping
2. **Command used to find hops:** tracert in Windows / traceroute in Linux
3. **Command to display IP configuration:** ipconfig in Windows / ip address in Linux
4. **Command to display TCP port status:** netstat
5. **Command to modify IP configuration in Linux:** ip address / ip route